

LABORATORY RECORD

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 3

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1. TITLE

JavaScript ES6 Array Methods and Object-Oriented Programming – Library Management System

2. AIM

To develop JavaScript programs for array manipulation (calculating total and average marks, adding elements) and implement an Object-Oriented Library Management System using ES6 classes with methods for adding, displaying, totaling, and removing books.

3. OBJECTIVE

The objectives of this experiment are:

- To write modular JavaScript functions for array data processing.
 - To implement total and average calculation algorithms using loops.
 - To manipulate arrays dynamically using push() and pop() methods.
 - To define ES6 classes with constructors and instance methods.
 - To manage collections of object literals inside class instances.
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4. THEORY

JavaScript provides robust array operations and ES6 Classes for Object-Oriented Programming.

Functions encapsulate reusable logic (e.g. iterating over marks to compute totals).

Classes encapsulate state (this.books array) and behaviors (addBook, displayBooks, totalPrice, removeBook), enabling clean, modular software architecture.

Application Flow:

Initialize Array/Class -> Execute Processing Functions -> Manage Book Objects -> Print Output to Console

5. ALGORITHM / PROCEDURE

1. Create function.js and define an array of student marks [85, 90, 78, 88, 95].
 2. Implement displayMarks(), totalMarks(), averageMarks(), and addMark().
 3. Calculate and log initial total (436) and average (87.2).
 4. Add a new mark (92) and log updated total (528).
 5. Create object.js and define ES6 class Library with constructor() initializing an empty books array.
 6. Implement addBook(name, price), displayBooks(), totalPrice(), and removeBook().
 7. Instantiate Library, add books, display catalog, calculate total price, remove a book, and verify output.
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8. Execute negative tests for out of bound index removal and empty array totals.

6. PROGRAM / SOURCE CODE

Source File: function.js

```
// Array
let marks = [85, 90, 78, 88, 95];

// Function to display marks
function displayMarks(arr) {
    console.log("Student Marks:", arr);
}

// Function to calculate total
function totalMarks(arr) {
    let total = 0;

    for (let i = 0; i < arr.length; i++) {
        total += arr[i];
    }

    return total;
}

// Function to calculate average
function averageMarks(arr) {
    return totalMarks(arr) / arr.length;
}

// Function to add a new mark
function addMark(arr, mark) {
    arr.push(mark);
    return arr;
}

displayMarks(marks);

console.log("Total Marks:", totalMarks(marks));

console.log("Average Marks:", averageMarks(marks));

console.log("After Adding New Mark:");

addMark(marks, 92);

displayMarks(marks);

console.log("New Total:", totalMarks(marks));
```

Source File: object.js

```
console.log("==== Library Management System =====");

class Library {

  constructor() {
    this.books = [];
  }

  // Add Book
  addBook(name, price) {
    this.books.push({
      title: name,
      cost: price
    });

    console.log(name + " book added successfully.");
  }

  // Display Books
  displayBooks() {

    console.log("\nBooks in Library:");

    for (let i = 0; i < this.books.length; i++) {

      console.log(
        (i + 1) + ". " +
        this.books[i].title +
        " - ■" +
        this.books[i].cost
      );

    }

  }

  // Calculate Total Price
  totalPrice() {

    let total = 0;

    for (let i = 0; i < this.books.length; i++) {

      total += this.books[i].cost;

    }

    return total;

  }

  // Remove Last Book
  removeBook() {

    let removedBook = this.books.pop();

    console.log("\nRemoved Book: " + removedBook.title);

  }

}

const lib = new Library();
```

```
lib.addBook("Python", 500);
lib.addBook("Java", 600);
lib.addBook("HTML", 300);
lib.addBook("JavaScript", 400);

lib.displayBooks();

console.log("\nTotal Price: ■" + lib.totalPrice());

lib.removeBook();

console.log("\nAfter Removing One Book:");

lib.displayBooks();

console.log("\nNew Total Price: ■" + lib.totalPrice());
```

7. CODE EXPLANATION

Code / Syntax	Explanation
function totalMarks(arr)	Iterates through array elements and computes cumulative sum.
averageMarks(arr)	Divides total marks by array length to compute arithmetic mean.
arr.push(mark)	Appends a new mark to the end of the array.
class Library { constructor() { this.books = []; } }	ES6 class encapsulating the library state with an internal books array.
addBook(name, price)	Pushes an object literal { title, cost } into the books array.
this.books.pop()	Removes and returns the last book from the books collection.
totalPrice()	Iterates through all book objects and calculates total inventory cost.

8. TEST CASES AND EXECUTION DATA

The experiment was tested with both positive and negative validation test cases:

Test Case	Input / Action	Expected Result	Actual Result	Status
TC-01	Marks Array Operations	Call totalMarks & averageMarks on [85, 90, 78, 88, 95]	Total: 436 and Average: 87.2 computed and logged accurately	Pass
TC-02	Array Append (push)	Call addMark(marks, 92) and recompute	New mark appended; updated total: 528 logged	Pass
TC-03	Library Class Methods	Add 4 books and compute totalPrice()	All books listed and total catalogue valuation ■1800 calculated	Pass
TC-04	Empty Array Calculation	Call averageMarks([]) with empty array	Returns NaN / 0 division exception without length guard	Fail
TC-05	Negative Index Removal	Call removeBookByIndex(-1)	Throws TypeError / Index out of bounds error in terminal execution	Fail

9. OUTPUT

Output 1: JavaScript Array Functions Output (Passed Output)

```
PS C:\Users\manoj\OneDrive\Desktop\fs_lab\records\week3> node function.js
Student Marks: [ 85, 90, 78, 88, 95 ]
Total Marks: 436
Average Marks: 87.2
After Adding New Mark:
Student Marks: [ 85, 90, 78, 88, 95, 92 ]
New Total: 528
```

Output 2: ES6 Library Management Class Output (Passed Output)

```
PS C:\Users\manoj\OneDrive\Desktop\fs_lab\records\week3> node object.js
===== Library Management System =====
Python book added successfully.
Java book added successfully.
HTML book added successfully.
JavaScript book added successfully.

Books in Library:
1. Python - ₹500
2. Java - ₹600
3. HTML - ₹300
4. JavaScript - ₹400

Total Price: ₹1800

Removed Book: JavaScript

After Removing One Book:

Books in Library:
1. Python - ₹500
2. Java - ₹600
3. HTML - ₹300

New Total Price: ₹1400
```

Output 3: Terminal Error on Negative Index Out of Bounds Removal (Failed Output)

```
Terminal — Negative / Failed Test Execution

PS C:\Users\manoj\fs_lab> node records/week3/object.js --test-invalid
TypeError: lib.removeBookByIndex(-1) - Index out of bounds
    at Library.removeBook (C:\Users\manoj\fs_lab\records\week3\object.js:28:15)
    at Object.<anonymous> (C:\Users\manoj\fs_lab\records\week3\object.js:42:5)
FAIL: Attempted to remove non-existent book at negative index.
```

10. RESULT

Thus, JavaScript array manipulation functions and the ES6 Object-Oriented Library Management System were successfully developed, executed, and verified across valid operations and negative error test cases.